

Name: Team Reef

Date: January 26, 2026

Class: Engineering

MyDesign Portfolio

Element A Initial Template

Useful Websites

- [Microplastic Fact Sheet](#)
- [Types of Microplastics: Primary & Secondary](#)
- [Microplastics info from NOAA](#)
- [Sampling and Analysis of Microplastic](#)

Additional resources (provided by Mary)

- [EPA's microplastic beach protocol](#)
- [Protocol and Data Collection Microplastics](#)
- [Collecting Methods for Microplastics](#)
- [Guidance on the monitoring of marine litter in European seas](#)

Research Articles:

- <https://link.springer.com/article/10.1007/s11356-020-07783-8>
- <https://www.sciencedirect.com/science/article/pii/S0048969724047090>
- <https://www.mdpi.com/2073-4441/12/4/1196>

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Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

Explain your problem in detail, including details that you might consider common knowledge. Use data and statistics to justify your problem statement. Do not provide a bulleted list of facts; rather, organize the problem details in paragraph form. Address as many of the 5 Ws +1 H (who, what, where, when, why, how) as is possible, for example:

- 1) Who is involved in this problem?
- 2) What are they doing?
- 3) What unique situations does the problem occur within, if any?
- 4) What is the size of the issue?
- 5) Where is the problem happening?
- 6) When does the problem occur?
- 7) Why does the problem happen?
- 8) How often does the problem happen?

Our problem is that microplastics in the ocean go uncollected and are creating extremely harsh problems for our environment. Mary, a scientist and teacher at our school, gave us the goal to build an ROV to collect data on microplastics in the Bay Area. She is going to Norway to present our research and our ROV design to One Ocean researchers and scientists. The main problem is that microplastics are sitting in the ocean and are often ingested by animals, specifically fish and birds. Microplastics can be found worldwide, on land and in water, from many sources, including landfills, residential households, construction projects, factories, agriculture, and more (Yale Sustainability). A microplastic is defined as a plastic particle measuring less than 5mm in size (Burton). Because they are so small in size, the microplastics can affect many parts of the ecosystem, such as fish and other marine life, as well as ocean swimmers. They also have a grave impact on all consumers, so in doing this research, we would be greatly benefiting these parts of our society. Our job is to find and create a way to collect samples of the water and microplastics.

Microplastics are a huge problem because they are everywhere, including the water we drink, the clothes we wear, and even the air we breathe, affecting all living beings' health. There have been multiple studies showing that microplastics have been found in organs

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and tissues, including the brain, heart, testicles, stomach, lymph nodes, and placenta (Savchuk). To date, microplastics have been found in over 1,300 species, and 114 of those species are marine species (Yvan). Over 100,000 marine mammals are killed by plastics every year (Yvan). Researchers have been noticing microplastics since 1970, but the term microplastics didn't come till 2004 (Savchuk). And since 1950, humans have produced over 8 billion tons of plastic (Yale Sustainability). 12700000 tons of plastic reach the ocean every year (Yvan).

Questions for Mary:

What organ are microplastics most harmful to?

Which organ or body system seems most vulnerable to microplastics?

What type of size of microplastics should be focused on?

How do researchers collect microplastic samples?

Why would an ROV be better than using scuba divers?

<https://news.stanford.edu/stories/2025/01/what-s-the-deal-with-microplastics-the-material-that-never-goes-away>

<https://sustainability.yale.edu/explainers/yale-experts-explain-microplastics>

<https://solve.mit.edu/solutions/66163>

<https://news.stanford.edu/stories/2025/09/microplastics-environment-human-health-impacts-research>

In order to earn a score of 5, you will need to cite at least five sources that are from the last ten years and that come from 3 or more of the following credible categories: peer reviewed journals, professional organizations, government agencies, local and national newspapers. Use in-line citations to explicitly state which information is from which source. Do not just list all of the citations at the end of the submission or in footnotes.

Additional questions to consider:

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- What are microplastics? What typical sizes are they?
- How do researchers currently collect water samples from the ocean?
- What kind of tools/hardware are capable of collecting sea water?
- What are the benefits to designing a robot to collect the water samples when a trained scuba diver can do the same work?
- How heavy is sea water? Will the weight/density limit how much water can be retrieved?
- How can sea water samples be stored and kept separate from one another?
- What are the ethical considerations of sampling sea water? How can we ensure marine life is not disturbed? Or at least the impact is minimized?
- Besides Mary, who would stand to benefit from what your team will design?
- How can we minimize contamination so we can ensure microplastics analyzed do actually come from the water sample?
- How many different kinds of micro plastics are there? Are we trying to analyze all of them? Some of them? Just one type?

Stakeholder Concerns and Connections (A3, A4)

Directly indicate at least five primary stakeholders or stakeholder groups. These are the people most directly impacted by the problem. Highlight the concerns of your primary stakeholders and use that to justify why there is a need to address the problem. Consider the breadth of the problem and how it might have social and environmental impacts on secondary stakeholders, which are people who may be impacted by a solution to the problem and are still part of the system.

Stakeholders:

- Mary
- Swimmers
- People who eat and drink
- Wildlife
- Marine Science Class

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Problem Importance (A6)

End Element A with a compelling argument for why this problem is important and needs to be solved now. Support this argument with data and statistics.

Our problem is that microplastics in the ocean go uncollected, creating severe environmental problems. Mary, a scientist and teacher at our school, gave us the goal to build an ROV to collect data on microplastics in the Bay Area. She is going to Norway to present our research and our ROV design to One Ocean researchers and scientists. The main problem is that microplastics are sitting in the ocean and are often ingested by animals, specifically fish and birds. Microplastics can be found worldwide, on land and in water, from many sources, including landfills, residential households, construction projects, factories, agriculture, and more (Yale Sustainability). A microplastic is defined as a plastic particle measuring less than 5mm in size (Burton). Because they are so small in size, the microplastics can affect many parts of the ecosystem, such as fish and other marine life, as well as ocean swimmers. They also have a grave impact on all consumers, so in doing this research, we would be greatly benefiting these parts of our society.

Microplastics are a huge problem because they are everywhere, including the water we drink, the clothes we wear, and even the air we breathe, affecting all living beings' health. There have been multiple studies showing that microplastics have been found in organs and tissues, including the brain, heart, testicles, stomach, lymph nodes, and placenta (Savchuk). To date, microplastics have been found in over 1,300 species, and 114 of those species are marine species (Yvan). Over 100,000 marine mammals are killed by plastics every year (Yvan). Researchers have been noticing microplastics since 1970, but the term microplastics didn't come till 2004 (Savchuk). And since 1950, humans have produced over 8 billion tons of plastic (Yale Sustainability). 12700000 tons of plastic reach the ocean every year (Yvan). Many people don't understand the severity of microplastics and how it negatively impacts their lives as well as the lives of marine life. Through designing an ROV and collecting water samples, we hope we can find a way to reduce microplastic waste.

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